Materials Science & Engineering A 724 (2018) 1–10

Contents lists available at ScienceDirect

Materials Science & Engineering A

journal homepage: www.elsevier.com/locate/msea

A study of the microstructures and mechanical properties of Ti6Al4V fabricated by SLM under vacuum

Bin Zhou[a,b,c], Jun Zhou[a,b,c], Hongxin Li[a,b,c], Feng Lin[a,b,c,*]

[a] *Department of Mechanical Engineering, Tsinghua University, Beijing 100084, China*
[b] *Biomanufacturing and Rapid Forming Technology Key Laboratory of Beijing, China*
[c] *Key Laboratory for Advanced Materials Processing Technology, Ministry of Education of China, China*

ARTICLE INFO

Keywords:
Selective laser melting
Vacuum
Ti6Al4V
HIP
X-ray CT

ABSTRACT

Ti6Al4V components with complex structures, excellent performance and good surface quality, can be fabricated by selective laser melting (SLM). The conventional SLM process is carried out in an inert gas filled chamber, at slightly greater pressure than ambient, to avoid oxidation at high temperature. However, the inert gas may cause pores in as-fabricated parts. In this study, the SLM process was carried out under vacuum in order to improve the quality of the SLM-fabricated Ti6Al4V samples. The as-fabricated Ti6Al4V samples then were subjected to hot-isostatic pressure (HIP). The remaining porosity was evaluated using X-ray computed tomography (CT). The microstructures and mechanical properties of the samples were evaluated with and without HIP. The test results showed that SLM under vacuum could reduce the porosity of Ti6Al4V samples, compared to material that was produced using the conventional SLM process. After HIP, porosity could be further reduced and the Ti6Al4V samples could achieve better elongation.

1. Introduction

Additive manufacturing technologies have attracted increasing attention because the products have many important applications due to their flexibility in design, possible complexity and the short lead times for production. The selective laser melting (SLM) process is a powder-based additive manufacturing technology, which utilizes a focused laser beam to melt the powder and produce parts, layer by layer, with high dimensional accuracy. Thus, SLM can be applied in the manufacture of impellers, blades, for porous structures and to produce medical implants [1–4].

The conventional SLM process is carried out in an inert gas filled chamber, at a pressure that is slightly greater than the ambient air pressure. Inert gas can prevent oxidation of the components by the high temperatures caused by the laser. However, the inert gas cannot melt into the lattice and, if not removed in time, the gas will be left inside the component, leading to porosity which can seriously affect the performance of SLM-fabricated parts [5–7]. While under vacuum, the volume of residual gas will be greatly reduced and therefore the quality of SLM-fabricated parts will be improved. Also, the metal spatter caused by gas expansion can be eliminated during laser irradiation [8,9]. However, the evaporation of volatile materials such as Al is much greater under vacuum and, in consequence, it is possible that the chemical composition of as-fabricated parts will change [10,11]. More significantly, the lenses of the laser will be 'contaminated' with metal vapor and, as a result, the SLM scanning process is detrimentally affected.

In the present investigation, SLM processing was carried out under vacuum in order to reduce the porosity of SLM-fabricated Ti6Al4V components. An anti-evaporation system was designed to prevent deposition of the metal vapor on the lenses under vacuum. The parameters for Ti6Al4V fabrication were studied. As-fabricated Ti6Al4V samples were treated using the hot-isostatic pressure (HIP) process to reduce residual porosity. The porosity values were evaluated using X-ray computed tomography (CT). The microstructures were characterized using scanning electron microscopy (SEM) and the mechanical properties were evaluated with and without HIP.

2. Methodology

2.1. Materials

Typical particles of gas-atomized spherical Ti6Al4V powder that was used in the SLM under vacuum process are shown in Fig. 1. The powder particle size was in the range of 13–55 µm, with an average of 32 µm, as measured using a laser particle size analyzer, Hydro 2000MU

* Corresponding author at: Department of Mechanical Engineering, Tsinghua University, Beijing 100084, China.
E-mail address: linfeng@tsinghua.edu.cn (F. Lin).

https://doi.org/10.1016/j.msea.2018.03.021
Received 13 January 2018; Received in revised form 4 March 2018; Accepted 5 March 2018
Available online 09 March 2018
0921-5093/ © 2018 Elsevier B.V. All rights reserved.

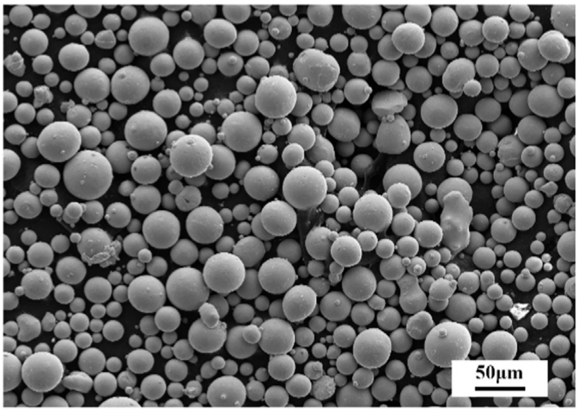

Fig. 1. SEM image of Ti6Al4V powders.

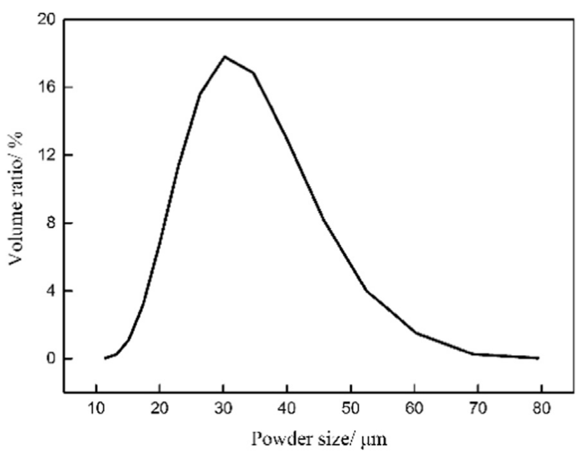

Fig. 2. Ti6Al4V powder size distribution.

Table 1
Composition of Ti6Al4V powder.

Element	Ti	Al	V	C	N	O
wt%	Balance	6.05	3.94	0.02	0.02	0.087

(A). The powder size distribution is shown in Fig. 2 and the composition is shown in Table 1.

2.2. Equipment

The tests were carried out in self-designed EBSM-250 electron beam selective melting unit at Tsinghua University (as shown in Fig. 3) [12]. In this unit, the laser beam is directed into the vacuum chamber through a sealed window with dynamic focusing scanning system that is located outside the vacuum chamber. An ytterbium fiber laser (YLR-200) with a maximum power of 200 W (IPG Laser GmbH, Germany) was used for the study.

The anti-evaporation system was designed to prevent the lenses (as shown in Fig. 4). There were two lenses in the anti-evaporation system, one was the sealed lens, which was in the chamber wall to maintain the vacuum pressure. The sealed lens needed to be absolutely clean or it would be very dangerous, so we used a second lens (the working lens) to protect the sealed lens. Behind the sealed lens, the working lens was placed, which was bigger. The working lens could rotate around its axis. The laser working area, through which the laser passes to the powder bed, was in one side of the cabinet and the brush was on the other side. Only the laser working area was exposed and would be contaminated. When the contaminated area rotated to the brush side, it would be cleaned. The as-designed system could allow the lens to work while being cleaned at the same time, and the brush did not disturb the SLM process. Fig. 5 shows the lens with (a) and without (b) anti-evaporation system after a long-time process. The anti-evaporation system indeed has a good use in the lens protection.

2.3. Processing parameter selection

The Ti6Al4V samples were fabricated using SLM under vacuum. The size of the samples was 20 mm × 20 mm × 10 mm, the chamber pressure was 1 Pa, the substrate was made of Ti6Al4V. The bulk energy density, E, [J/mm^3] was introduced to explain the effect of process parameters, referred to the energy that the powder obtained in unit volume when the laser melted each layer. The bulk energy density formula was:

$$E = \frac{P}{hvL} \tag{1}$$

where P is the laser power [W], v is the scanning speed [mm/s]; h is the hatching space [mm] and L is the layer thickness [mm].

In the SLM under vacuum process, the powder thickness $L = 0.05$ mm. The powder bed temperature was 200 °C. The range of forming parameters for Ti6Al4V fabricated using SLM under vacuum in the previous experiments is shown in Fig. 6, with a hatching pace $h = 0.05$ mm. When $E > 400$ J/mm^3, the Ti6Al4V was formable. Ti6Al4V samples fabricated using SLM under vacuum are shown in Fig. 7, and the associated parameters are presented in Table 2. Due to the limitation of our equipment, the comparison samples produced under positive pressure (conventional SLM environment) were cited from the literatures.

2.4. Characterization

The morphologies and microstructures of the Ti6Al4V samples were analyzed using a scanning electron microscope (SEM, ZEISS-SUPRA55), operating at 15 kV. The samples were etched for 15 s with Kroll's reagent: 1 mL HNO_3, 2 mL HF and 97 mL of distilled water. The chemical compositions of the powder and samples were measured using scanning electron microscope - energy dispersive spectroscopy (SEM-EDS, X-Max, Oxford Instruments) with a resolution of 125 eV (Mn Kα). X-ray diffraction (XRD) was carried out to evaluate the differences in phase composition. The range of diffraction angle 2θ was 30°–90° and the scanning speed was 10°/min with the pace of 0.02°. The densities of the Ti6Al4V samples were measured by the method of drainage, using a digital instrument (ADVENTURER, AR423DCN) with an accuracy of 0.001 g. Tensile samples, shown in Fig. 8, were tested using a tensile test machine (WDW-100/E) with a maximum load of 100 kN. The loading rate was 0.05 mm/min. Samples IV-VI were treated with the hot-isostatic pressure (HIP) process at 100 MPa, 920 °C for 2 h in an argon atmosphere. After the HIP treatment, Sample V was heat-treated at 900 °C for 2 h under vacuum. The remaining porosity in the samples was evaluated by X-ray computed tomography (CT, nanoVoxel-4000), operating at 100 kV, 10 W and with a resolution of 3.96 μm.

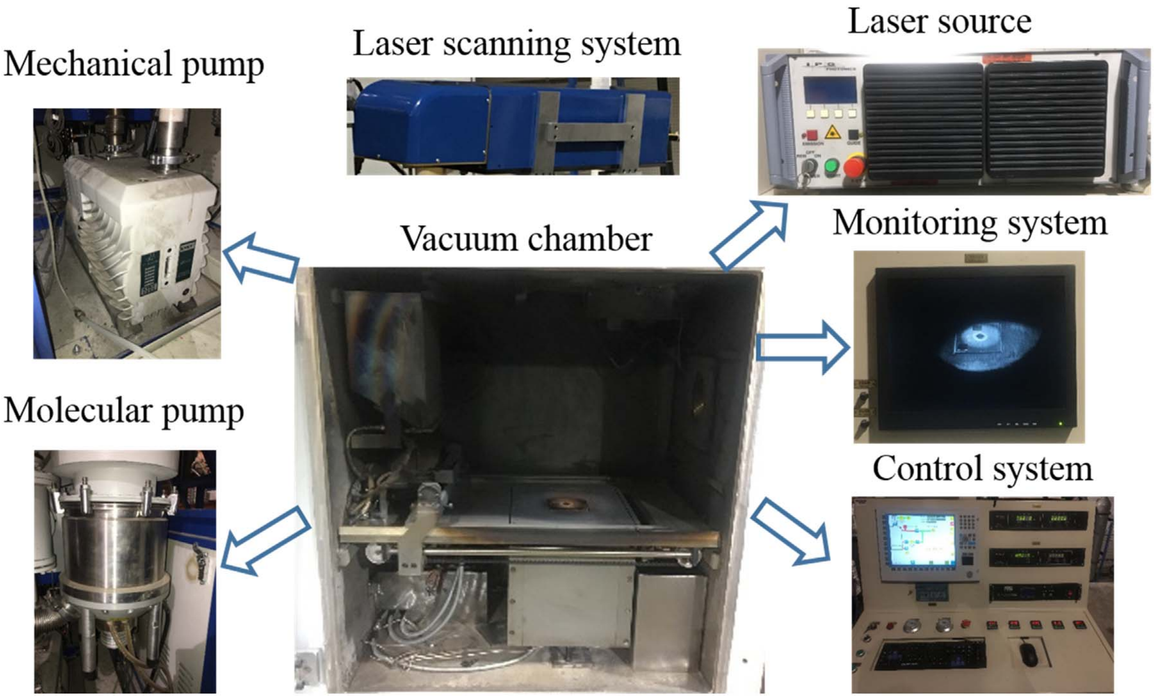

Fig. 3. The Equipment of the SLM process under vacuum.

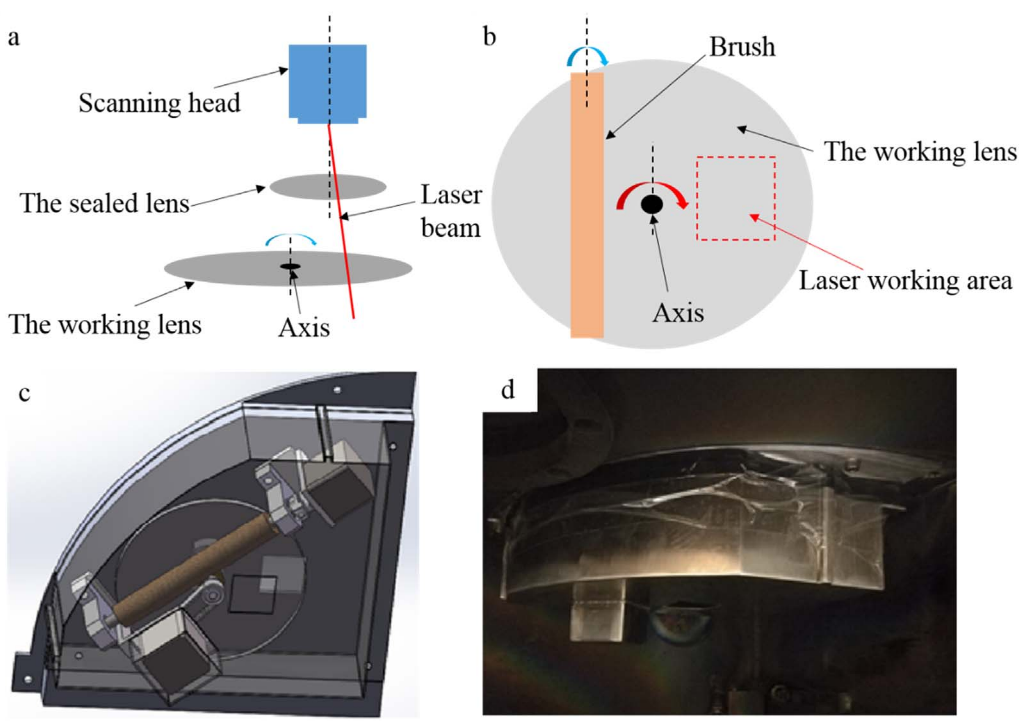

Fig. 4. Anti-evaporation system.

3. Results

3.1. Microstructures

Bulk energy density E has a significant influence on the fine acicular martensite and the lamellar α phase. With the increase of bulk energy density, the heat obtained by molten pool increases, and the molten pool becomes deeper and wider. The columnar crystal coarsening occurs when bulk energy density increases. The greater the energy density is, the higher temperature the metastable fine acicular will get in the reheat cycle. The fine acicular martensite fully changes to a Widmanstatten structure, and elongated lamellar α phase also will

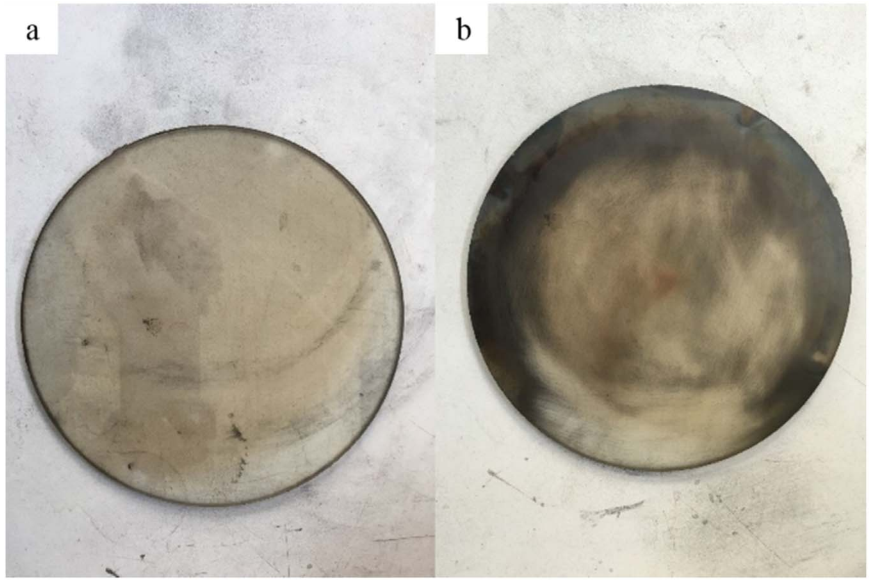

Fig. 5. The lens with (a) and without (b) anti-evaporation system after a long-time process.

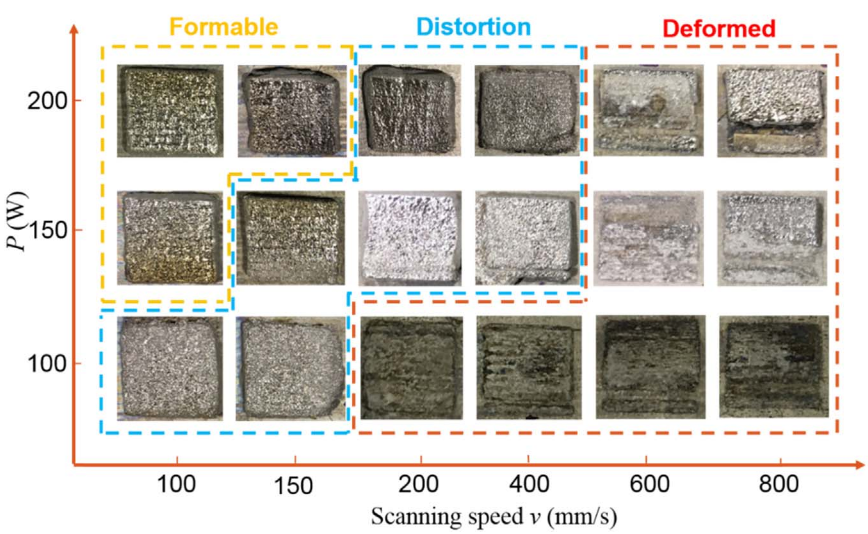

Fig. 6. Formable parameter range of Ti6Al4V fabricated using SLM under vacuum.

coarsen [13–16].

(1) When the bulk energy density $E < 533\,J/mm^3$, the energy density is too low, which leads to the partial melting of the powder and the temperature gradient in the molten pool is more uniform, forming equiaxed grains (Fig. 9(a)).
(2) When the bulk energy density $E = 600–1333\,J/mm^3$, the temperature gradient along the building direction is larger than other directions and fine acicular martensite grows paralleled in the grains. In the original β phase, α lamellae form in different orientations and precipitate continuously at β grain boundaries. With an increase in bulk energy density, α lamellae become coarser. The original β grain boundaries are destroyed and form a Widmanstatten structure (Fig. 9(b)-(f));

SEM images of Ti6Al4V samples produced by SLM under vacuum after HIP treatment are shown in Fig. 10. After HIP treatment at 920 °C for 2 h, α lamellae grew significantly coarser, compared to the as-fabricated samples. The coarsened α phase after the HIP treatment was attributed to the high heat treatment temperature. When heated, the α phase is nucleated along the α′ boundaries and vanadium atoms are expelled, leading to the formation of β at α phase boundaries [17,18]. Hence during HIP treatment, the long α′ needles are transformed into the long and thick α lamellae [4].

3.2. Phase analysis and chemical composition

The XRD spectra of the Ti6Al4V samples are shown in Fig. 11. For all the six samples, most of the peaks could be identified as α/α′ phase. As α phase and α′ phase have the same hexagonal close-packed (hcp) structure and a similar lattice parameter, it is difficult to differentiate the two phases [6].

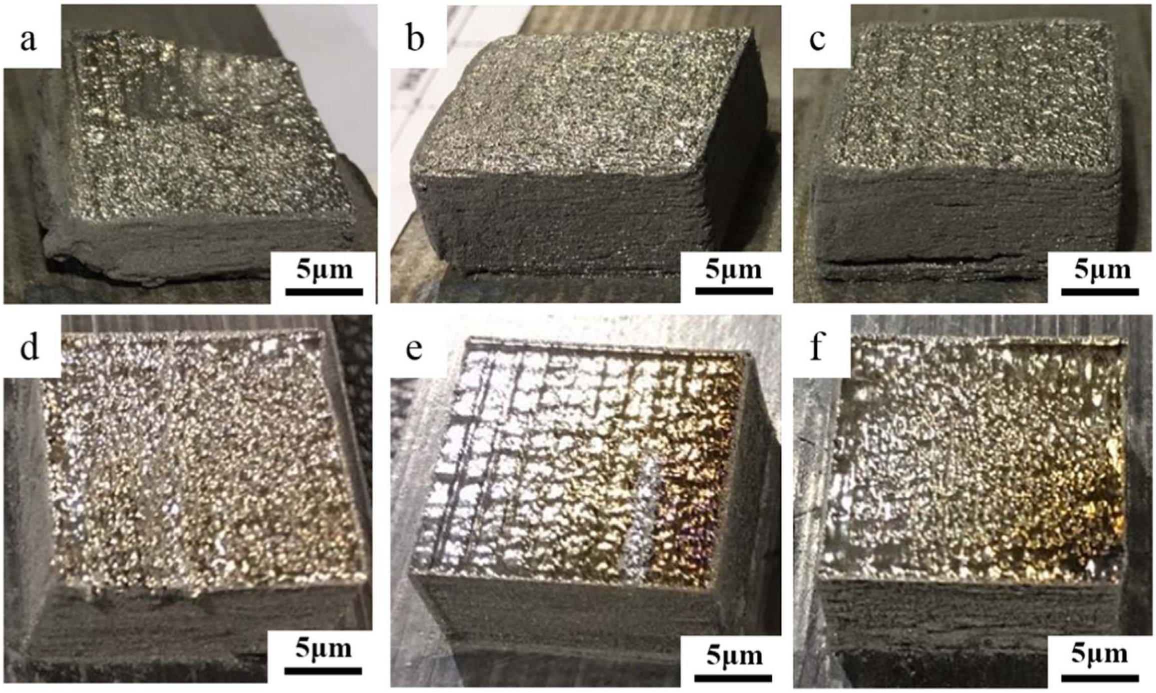

Fig. 7. Ti6Al4V samples fabricated using SLM under vacuum: (a) Sample I; (b) Sample II; (c) Sample III; (d) Sample IV; (e) Sample V; (f) Sample VI.

Table 2
Parameters of the Ti6Al4V samples.

	Sample I	Sample II	Sample III	Sample IV	Sample V	Sample VI
P(W)	200	150	200	200	150	200
v(mm/s)	150	100	100	150	100	100
h(mm)	0.05	0.05	0.05	0.03	0.03	0.03
E(J/mm^3)	533	600	800	889	1000	1333

P-Laser power; *v*-scanning speed; *h*-hatching space; *E*-bulk energy density.

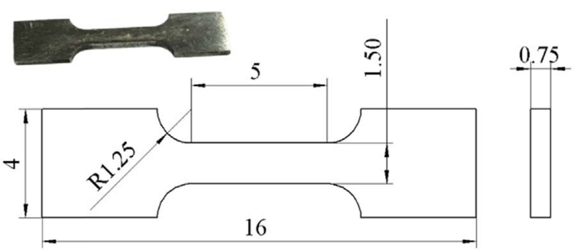

Fig. 8. Tensile samples.

The SEM-EDS results are shown in Fig. 12. In the Ti6Al4V samples, the contents of Ti, Al and V were slightly lower than that of the original Ti6Al4V powder. There were no obvious compositional differences after SLM under vacuum process.

3.3. Porosity and density

The density results for the Ti6Al4V samples are shown in Fig. 13, as measured by the drainage method. As a result, the higher energy input contributed to the higher density. The density of Sample V (E = 1000 J/mm^3) reached a maximum of 4.37 g/cm^3.

The X-ray CT reconstruction of Sample V is shown in Fig. 14. The middle parallel part of tensile sample V was reconstructed by X-ray CT. Many micron pores existed in Sample V (as shown in Fig. 15(a)). The porosity of the as-fabricated Sample V (~0.095%) was lower than that of the conventional SLM samples (~ 0.290%) [4]. After HIP treatment, the number of pores was reduced significantly, especially pores with small size (as shown in Fig. 15(a) and (b)) and the porosity of Sample V was reduced to 0.067%. With heat treatment after HIP, the porosity (~ 0.044%) was further reduced.

3.4. Tensile properties

Stress-strain curves for the Ti6Al4V samples are shown in Fig. 16. With the increase of energy density, the tensile strength of the Ti6Al4V samples produced by SLM under vacuum increased. When E = 1000 J/mm^3, the ultimate tensile strength increase reached a maximum of 1008 MPa. But when E = 1333 J/mm^3, the energy density was too high and the molten pool temperature became higher, so that α lamellae were coarser in the Widmanstatten structure (Fig. 9(e)-(f)), which

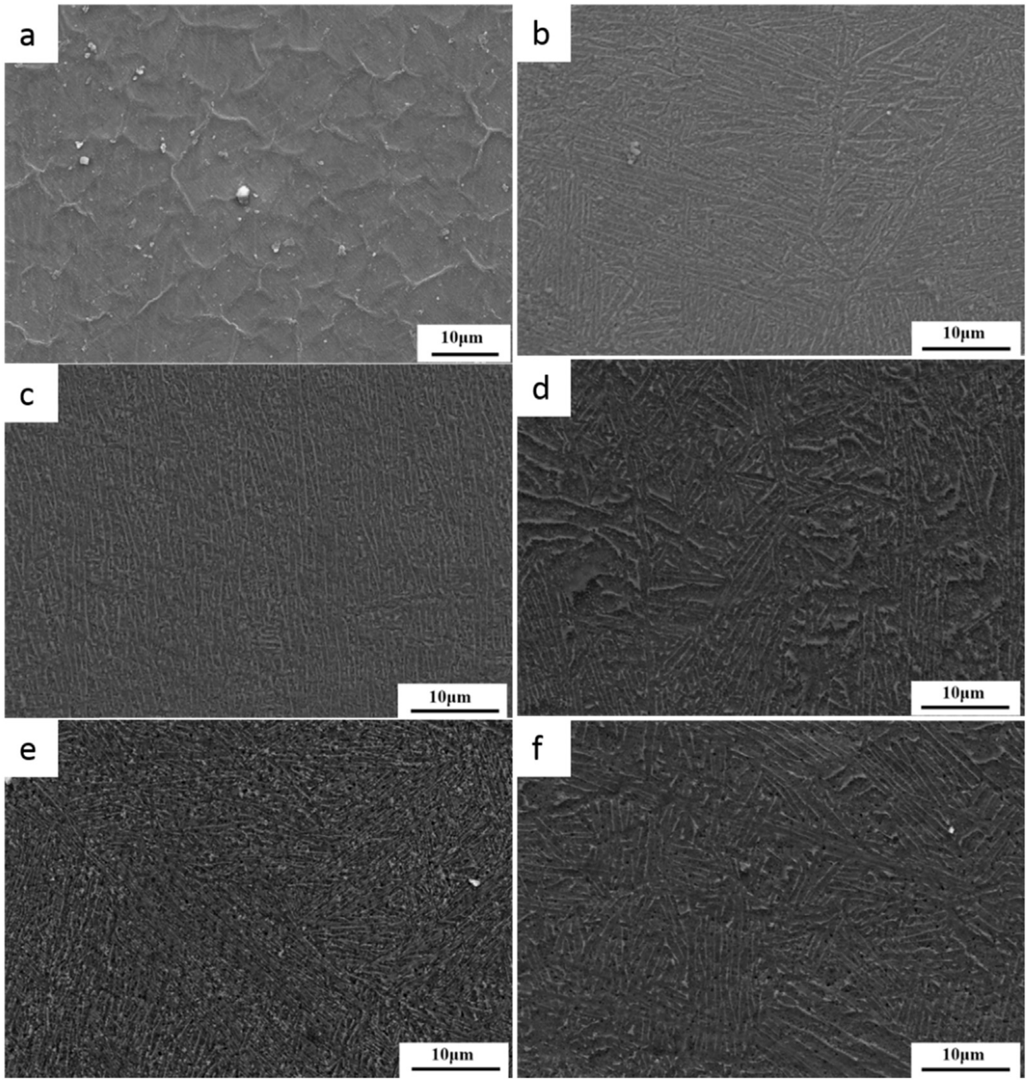

Fig. 9. SEM images of Ti6Al4V samples fabricated by SLM under vacuum : (a) Sample I; (b) Sample II; (c) Sample III; (d) Sample IV; (e) Sample V; (f) Sample VI.

resulted in a lower tensile strength.

Simonelli [19] achieved the ultimate tensile strength of 1117 MPa and an elongation of 8.9% of the SLM Ti6Al4V samples, using a Renishaw AM250 with P (157 W), v (225 mm/s), and h (0.1 mm). While Zhao [4] used a Realizer PBF-L to fabricate Ti6Al4V samples with P (200 W), v (1000 mm/s), and h (0.1 mm), giving an ultimate tensile strength of 1300 MPa and an elongation of 6%. Compared to the above results from the conventional SLM process, the Ti6Al4V samples fabricated by SLM under vacuum had a relatively lower ultimate tensile strength but had a better elongation.

Sample V with HIP treatment achieved a larger elongation but had a lower tensile strength compared to the same sample without HIP. The difference in the tensile results was due to the differences observed in the microstructures. The lower tensile strength and larger elongation observed in Sample V with HIP treatment can be attributed to the coarser α lamellae in the microstructure (Fig. 10(b)). After HIP treatment, the porosity of the samples was reduced significantly, which played an important role in the improvement in the elongation performance.

4. Discussion

4.1. Energy density and chamber pressure

According to the results shown in Fig. 7, the formable input energy density range of Ti6Al4V samples fabricated using SLM under vacuum was 800–1333 J/mm^3, with a chamber pressure of 1 Pa, whereas in the conventional SLM process, 120 J/mm^3 is sufficient to manufacture Ti6Al4V alloy with full density [20]. In order to investigate the relationship of energy density and vacuum degree, extra experiments were designed and carried out in which the Ti6Al4V samples were fabricated with chamber pressures from 10^{-3} Pa to 10^5 Pa.

The results of tests on the Ti6Al4V samples that were fabricated with different chamber pressures are summarized in Fig. 17. With a decrease in chamber pressure (i.e. with an increase in the degree of vacuum), the formable energy input increased significantly. Heat transfer between the powder particles requires a medium, which turns out to be air, to be present during the SLM process. Less air in the chamber makes heat transfer more difficult between the powder

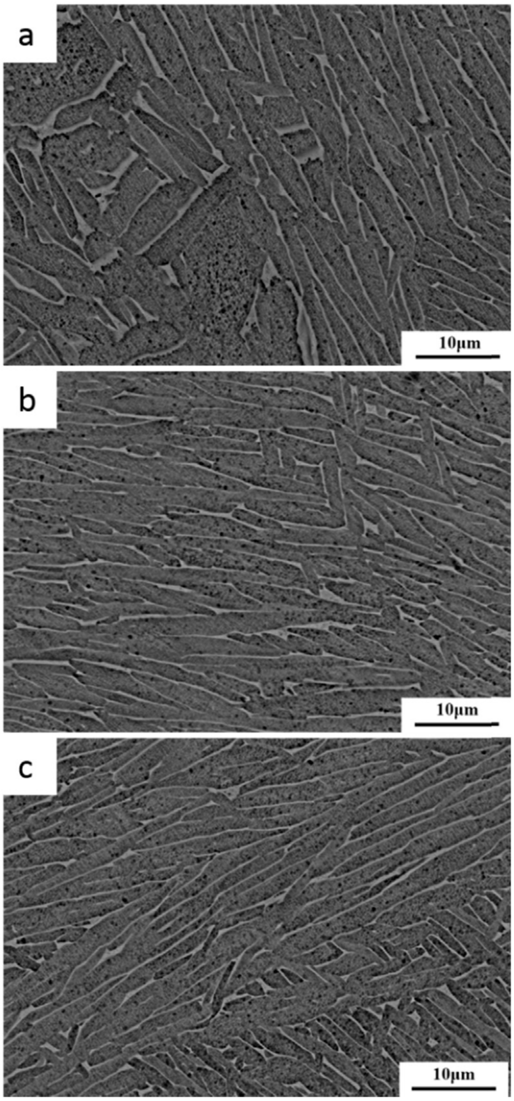

Fig. 10. SEM images of Ti6Al4V samples after HIP treatment: (a) Sample IV; (b) Sample V; (c) Sample VI.

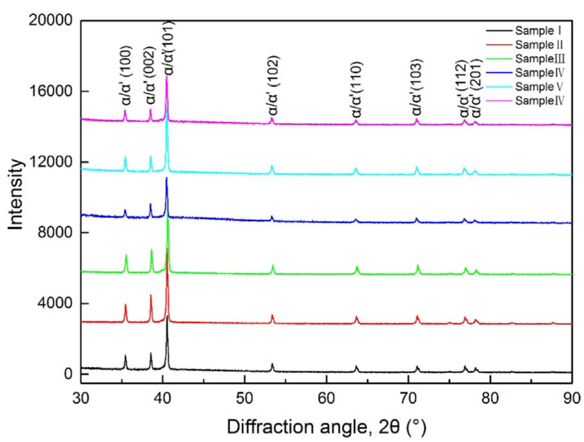

Fig. 11. XRD spectra of Ti6Al4V samples.

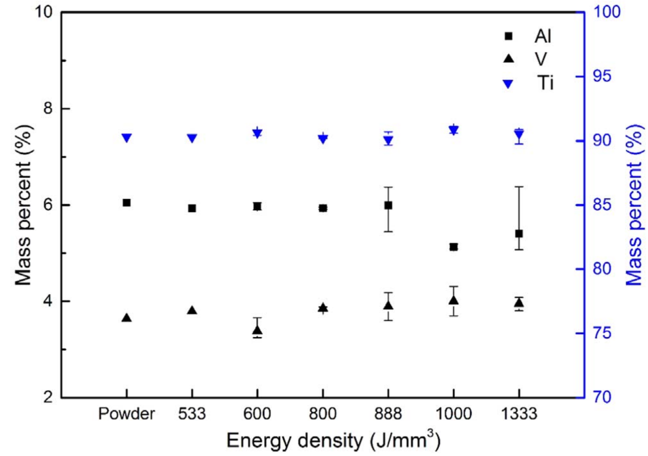

Fig. 12. Chemical composition of Ti6Al4V samples.

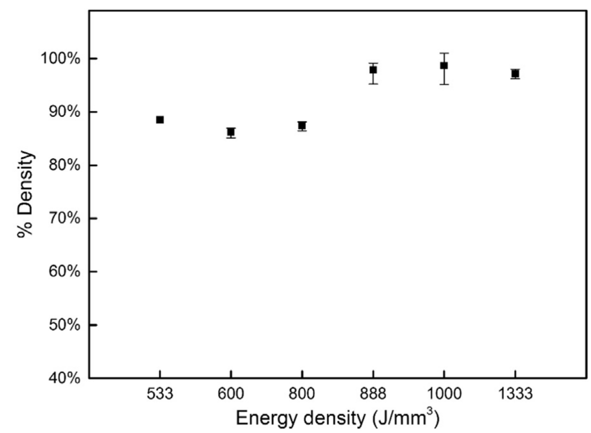

Fig. 13. Density of Ti6Al4V samples.

particles. When the chamber pressure is decreased, more energy input is needed to form a continuous metal molten pool and eventually to get a smooth and even surface in each layer.

The boiling temperatures of metal element (i.e. Ti) will decrease under vacuum, which can be calculated by the Clausius-Clapeyron relation [11], shown in Eq. (2).

$$\ln\left(\frac{P_2}{P_1}\right) = \frac{\Delta H_v}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \tag{2}$$

where ΔH_v is the enthalpy change of Ti from liquid state to gas state.ΔH_v = 431.142 kJ/mol and R = 8.314 J/(mol K). Under standard condition, $P_1 = 10^5$ Pa, T_1 = 3637 K, so we can get the results of the boiling temperatures of Ti under different pressures (as shown in Fig. 18). During the SLM under vacuum process, the metal vapor was much more than that in conventional SLM process. The metal vapor will be ionized as plasma by laser on top of the molten pool. The plasma shield and metal vapor can absorb laser energy and defocus the laser beam, which cause the increase of input energy, as shown in Fig. 17. The plasma shield and metal vapor will float around in chamber eventually be pumped away.

When the chamber pressure was 10^5 Pa, which is near the pressure of the conventional SLM process, the energy density for formable fabrications (380–570 J/mm^3) was a little higher, compared to the requirement for the conventional SLM process. In the bespoke equipment

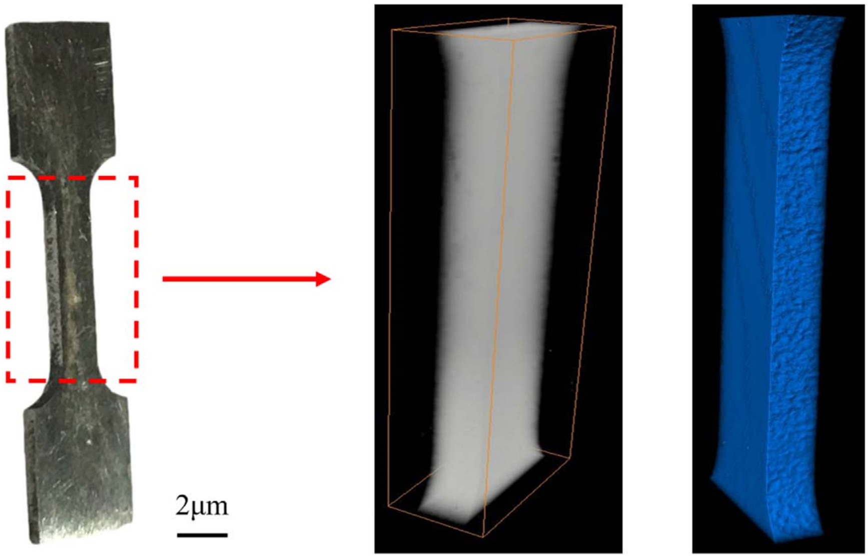

Fig. 14. X-ray CT reconstruction of Sample V.

developed for the present project, the laser needed to pass through two lenses, a seal lens and the working lens (as shown in Fig. 4). Each of the lenses has a certain absorption of the laser beam. Although the brush of the anti-evaporation cleaning system can remove deposited metallic deposition from the working lens, after longer periods of use, the working lens can become 'contaminated with a thin residual layer of vaporized metal, which increases the degree of absorption of the laser beam.

More seriously, in the bespoke unit, the laser was not placed in the axial direction of the powder bed. As the laser beam irradiates its power with an angle of 8° against the axial direction, the beam spot on the powder was an ellipse, not a circle, and the energy distribution was more scattered. In consequence, the energy density must be increased in order to melt the powder.

4.2. HIP improvement of SLM under vacuum

It is known from other research [21–23] that the defects in the SLM samples were mainly due to lack of fusion voids and pores. In the conventional SLM process, researchers have suggested three main mechanisms by which these pores are produced. First, when SLM processes are operated at very high energy density, deposition or melting may be performed in keyhole mode [24]. Without careful control of keyhole mode melting, the keyholes can become unstable and repeatedly form and collapse, leaving voids inside the deposit that contain entrapped metal vapor [25]. Second, the inert gas can be entrapped inside the powder particles during the powder atomization process. Third, pores may be formed due to the entrapment of the inerting gas in the chamber. HIP treatment can minimize or eliminate defects due to lack of fusion and keyhole pores. However, as the inerting gas cannot be dissolved into the α-Ti lattice, the inerting gas pores are just slightly reduced in size but cannot be completely eliminated by HIP treatment. Furthermore, when the 'HIPed' samples were heated, the inert gas pores can regrow [26] and the porosity, therefore, will increase. Williams [27] reported that argon gas pores of the HIPed Ti6Al4V samples had been found to reappear and grow progressively in proportion to their original (as-fabricated) size after the high temperature treatment. Thus, the HIP treatment can deliver limited improvements. This is one of the most important reasons that HIP treatment is not necessary for modified SLM process.

From the results shown in Fig. 15, the porosity of the Ti6Al4V samples was reduced significantly after HIP treatment and, after a 2 h heat treatment at 900 °C, the porosity was further reduced. In the SLM under vacuum process, the defects comprised mainly of lack of fusion voids and pores. However, the pores consisted mainly of vacuum pores, because of the vacuum chamber environment. After HIP treatment, the vacuum pores mostly were eliminated. Thus, the porosity of the Ti6Al4V samples could be reduced significantly. When the samples with HIP treatment were heated, only the inert pores with entrapped gas inside the powder particles get larger. The vacuum pores and lack of fusion voids could be further minimized and thus, the porosity of samples with HIP was further reduced after heat treatment.

Form the results shown in Fig. 16, samples that had been fabricated by SLM under vacuum and subjected to the HIP treatment could achieve larger elongations (nearly 16%). Qian [26] reported that HIP treatment would increase the elongation by more than 10% for conventional SLM Ti6Al4V samples, which gives some improvements, but they are limited. According to the above results, Ti6Al4V samples that have been fabricated by the SLM under vacuum process with HIP can have reduced porosity and better elongation, compared to samples produced using the conventional SLM process.

5. Conclusions

SLM process was carried out under vacuum and Ti6Al4V samples were fabricated in a bespoke process unit. The conclusions can be afforded as following:

(1) The anti-evaporation system was designed to prevent the lenses from the metal vapor and indeed has a good use in the lens protection.
(2) The microstructures of samples mainly were composed of fine acicular martensite, and Widmanstatten, consisting of α/α′ phase. There were no obvious compositional differences after the SLM under vacuum process.
(3) The porosity of the as-fabricated samples was lower than that of the

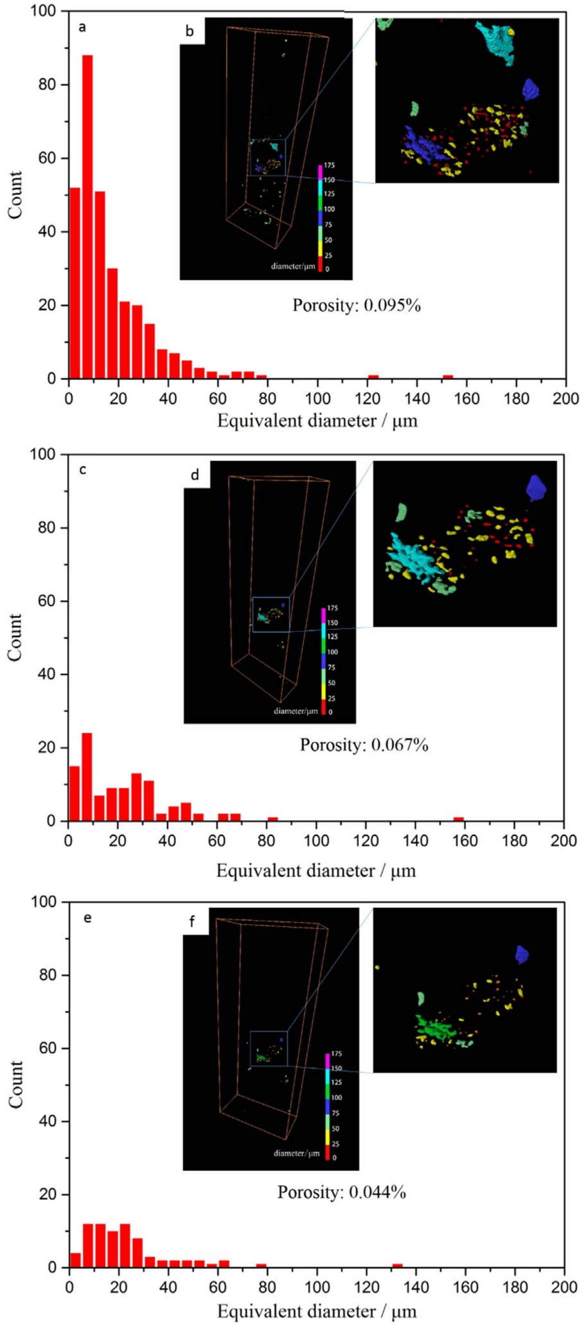

Fig. 15. X-ray CT-scan results ((b), (d) and (f)) and pore distribution ((a), (c) and (e)) of Sample V; (a) and (b) as-fabricated; (c) and (d) HIP; (e) and (f) heat treatment after HIP.

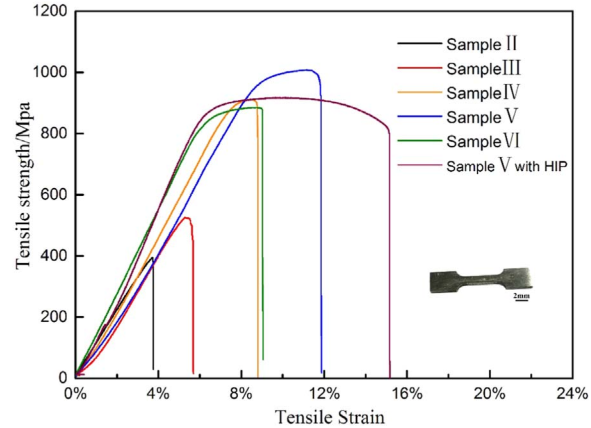

Fig. 16. Stress-strain curves of the Ti6Al4V samples.

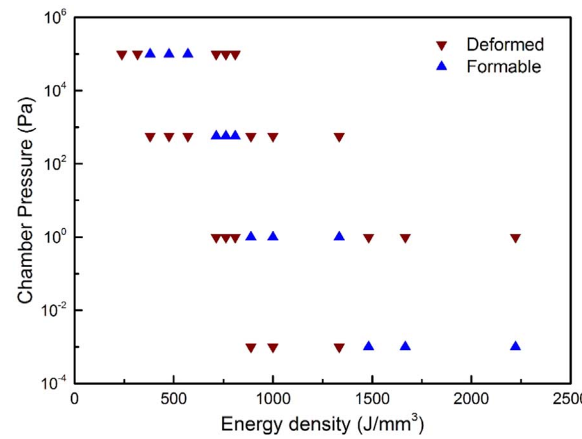

Fig. 17. Ti6Al4V samples fabricated at different chamber pressures.

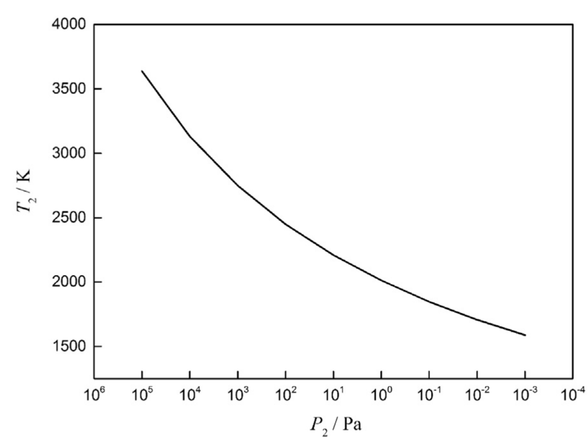

Fig. 18. The boiling temperatures of Ti under different pressures.

conventional SLM samples, and HIP post-processing could reduce the porosity and achieve better elongation of the as-fabricated samples than the conventional SLM samples with HIP. The porosity of samples with HIP was further reduced after heat treatment, which made HIP a necessary process to post-treat the parts using SLM under vacuum process.

(4) With decreasing chamber pressure (i.e. with an increase of the degree of vacuum), the formable energy input increased significantly. The formable input energy density range of Ti6Al4V samples fabricated using SLM under vacuum was 800–1333 J/mm^3 with a chamber pressure of 1 Pa.

Acknowledgements

This work was supported by the 'Suzhou-Tsinghua Special Innovation Leading Action' program 2016SZ0216.

References

[1] D. Agius, K.I. Kourousis, C. Wallbrink, T. Song, Cyclic plasticity and microstructure of as-built SLM Ti-6Al-4V: the effect of build orientation, Mater. Sci. Eng. A 701 (2017) 85–100.
[2] W.J. Ge, C. Guo, F. Lin, Microstructures of components synthesized via electron beam selective melting using blended pre-alloyed powders of Ti6Al4V and Ti45Al7Nb, Rare Met. Mater. Eng. 44 (2015) 2623–2627.
[3] J.P. Kruth, L. Froyen, J. Van Vaerenbergh, P. Mercelis, M. Rombouts, B. Lauwers, Selective laser melting of iron-based powder, J. Mater. Process. Technol. 149 (2004) 616–622.
[4] X. Zhao, S. Li, M. Zhang, Y. Liu, T.B. Sercombe, S. Wang, Y. Hao, R. Yang, L.E. Murr, Comparison of the microstructures and mechanical properties of Ti-6Al-4V fabricated by selective laser melting and electron beam melting, Mater. Des. 95 (2016) 21–31.
[5] L.E. Murr, S.M. Gaytan, A. Ceylan, E. Martinez, J.L. Martinez, D.H. Hernandez, B.I. Machado, D.A. Ramirez, F. Medina, S. Collins, Characterization of titanium aluminide alloy components fabricated by additive manufacturing using electron beam melting, Acta Mater. 58 (2010) 1887–1894.
[6] D. Gu, Y. Hagedorn, W. Meiners, G. Meng, R.J.S. Batista, K. Wissenbach, R. Poprawe, Densification behavior, microstructure evolution, and wear performance of selective laser melting processed commercially pure titanium, Acta Mater. 60 (2012) 3849–3860.
[7] J. Sun, Y. Yang, D. Wang, Parametric optimization of selective laser melting for forming Ti6Al4V samples by Taguchi method, Opt. Laser Technol. 49 (2013) 118–124.
[8] H. Schleifenbaum, W. Meiners, K. Wissenbach, C. Hinke, Individualized production by means of high power selective laser melting, CIRP J. Manuf. Sci. Technol. 2 (2010) 161–169.
[9] D. Buchbinder, H. Schleifenbaum, S. Heidrich, W. Meiners, J. Bültmann, High power selective laser melting (HP SLM) of aluminum parts, Phys. Procedia 12 (2011) 271–278.
[10] I. Yadroitsev, P. Bertrand, I. Smurov, Parametric analysis of the selective laser melting process, Appl. Surf. Sci. 253 (2007) 8064–8069.
[11] B. Zhang, H. Liao, C. Coddet, Selective laser melting commercially pure Ti under vacuum, Vacuum 95 (2013) 25–29.
[12] C. Guo, J. Zhang, J. Zhang, W. Ge, B. Yao, F. Lin, Scanning system development and digital beam control method for electron beam selective melting, Rapid Prototyp. J. 21 (2015) 313–321.
[13] L.E. Murr, E.V. Esquivel, S.A. Quinones, S.M. Gaytan, M.I. Lopez, E.Y. Martinez, F. Medina, D.H. Hernandez, E. Martinez, J.L. Martinez, S.W. Stafford, D.K. Brown, T. Hoppe, W. Meyers, U. Lindhe, R.B. Wicker, Microstructures and mechanical properties of electron beam-rapid manufactured Ti-6Al-4V biomedical prototypes compared to wrought Ti-6Al-4V, Mater. Charact. 60 (2009) 96–105.
[14] L. Thijs, F. Verhaeghe, T. Craeghs, J.V. Humbeeck, J. Kruth, A study of the microstructural evolution during selective laser melting of Ti-6Al-4V, Acta Mater. 58 (2010) 3303–3312.
[15] T. Vilaro, C. Colin, J.D. Bartout, As-Fabricated and Heat-Treated Microstructures of the Ti-6Al-4V Alloy Processed by Selective Laser Melting, Metall. Mater. Trans. A 42 (2011) 3190–3199.
[16] W.Q. Toh, P. Wang, X. Tan, M. Nai, E. Liu, S. Tor, Microstructure and wear properties of electron beam melted Ti-6Al-4V parts: a comparison study against as-cast form, Metals 6 (2016) 284–297.
[17] F.X. Gil Mur, D. Rodríguez, J.A. Planell, Influence of tempering temperature and time on the α′-Ti-6A1–4 V martensite, J. Alloy. Compd. 234 (1996) 287–289.
[18] B. Vrancken, L. Thijs, J. Kruth, J. Van Humbeeck, Heat treatment of Ti6Al4V produced by selective laser melting: microstructure and mechanical properties, J. Alloy. Compd. 541 (2012) 177–185.
[19] M. Simonelli, Y.Y. Tse, C. Tuck, Effect of the build orientation on the mechanical properties and fracture modes of SLM Ti-6Al-4V, Mater. Sci. Eng. A 616 (2014) 1–11.
[20] L. Zhang, H. Attar, Selective laser melting of titanium alloys and titanium matrix composites for biomedical applications: a review, Adv. Eng. Mater. 18 (2016) 463–475.
[21] R. Morgan, C.J. Sutcliffe, W. O'Neill, Density analysis of direct metal laser re-melted 316L stainless steel cubic primitives, J. Mater. Sci. 39 (2004) 1195–1205.
[22] E.O. Olakanmi, R.F. Cochrane, K.W. Dalgarno, Densification mechanism and microstructural evolution in selective laser sintering of Al-12Si powders, J. Mater. Process. Technol. 211 (2011) 113–121.
[23] Q. Jia, D. Gu, Selective laser melting additive manufacturing of Inconel 718 superalloy parts: densification, microstructure and properties, J. Alloy. Compd. 585 (2014) 713–721.
[24] W.E. King, H.D. Barth, V.M. Castillo, G.F. Gallegos, J.W. Gibbs, D.E. Hahn, C. Kamath, A.M. Rubenchik, Observation of keyhole-mode laser melting in laser powder-bed fusion additive manufacturing, J. Mater. Process. Technol. 214 (2014) 2915–2925.
[25] A. Kaplan, A model of deep penetration laser welding based on calculation of the keyhole profile, J. Phys. D: Appl. Phys. 27 (1994) 1805–1814.
[26] M. Qian, W. Xu, M. Brandt, H.P. Tang, Additive manufacturing and postprocessing of Ti-6Al-4V for superior mechanical properties, MRS Bull. 41 (2016) 775–784.
[27] S. Tammas-Williams, P.J. Withers, I. Todd, P.B. Prangnell, Porosity regrowth during heat treatment of hot isostatically pressed additively manufactured titanium components, Scr. Mater. 122 (2016) 72–76.